

Module 1

- Explain Security Goals.
- Define Active and Passive Attacks.

- Discuss Goals of Security.

- Explain Euclidean algorithm.

- Explain Substitution and Transposition techniques.

- Difference between Monoalphabetic and Polyalphabetic substitution techniques.

- Explain Vigenere cipher.

- Example on playfair cipher, Hill cipher

- Explain with examples Keyed and keyless transposition ciphers.

Notebook

Module 2

- Explain Data Encryption Standard (DES) algorithm.
- Explain AES algorithm.
- Compare AES & DES.
- Explain ECB and CBC modes of block cipher.
- Explain Principles of Block Cipher.
- Explain Cipher feedback and Output feedback mode of block cipher.
- Explain RC4 algorithm
- Any example on RSA algorithm & diffie hellman key exchange algo.
- Explain digital certificate X.509.
- Explain Kerberos.

Module 3

- List properties of Secure Hash Function.
- Compare SHA-1 and MD5
- List types of Authentication. Explain any one.
- What is need for message authentication? List various techniques used for message authentication. Explain any one.
- Distinguish between HMAC and CMAC.
- Discuss Message Authentication protocol(MAC).
 - Explain MD5 algorithm in detail.

Module 4

- Explain digital signature.
- Discuss various attacks on Digital Signatures.

Module 5

- Explain SQL injection Attack.
- Write a note on: ICMP flood attack.
- Explain Denial of service (DOS) attack.
- write a short note on: Packet Sniffing.
- Explain ARP Spoofing
- Explain PGP.
- Explain IPSEC.

Module 6

- Write short note on: Buffer Overflow.
- Define Worms and Viruses.
- List types of malicious program.
- List Types of computer viruses.